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Electrochemical battery module and method therefor

Abstract

A method for assembling a battery cell includes assembling an electrode stack that includes first electrodes, second electrodes, and separators. A first conductive adhesive pad is formed on an end cap, wherein the first conductive adhesive pad electrically couples to a first terminal arranged on the end cap. A second conductive adhesive pad is formed on an enclosure, wherein the second conductive adhesive pad electrically couples to a second terminal arranged on the enclosure. The end cap is installed onto the electrode stack with the first cell tabs engaging the first conductive adhesive pad. The electrode stack is inserted into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad. The end cap is assembled onto the enclosure.

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Background/Summary

INTRODUCTION
(1) The disclosure relates to an electrochemical battery module, and a method of fabricating the battery module.
(2) Battery modules may be constructed from a plurality of electrochemical cells that are electrically interconnected. Each of the electrochemical cells includes a positive terminal and a negative terminal, with selected terminals of each of the electrochemical cells being soldered or welded to mechanically attach and electrically connect the cells to positive and negative terminals of the battery module.
(3) There is a need to provide an improved battery module and an improved process for mechanically and electrically joining elements of a battery module.
SUMMARY
(4) The concepts described herein provide a battery module having positive and negative terminals that are electrically connected and weldlessly, mechanically joined to an electrode stack employing an electrically conductive mechanically adhesive material.
(5) An aspect of the disclosure may include a method for assembling a battery cell that includes assembling an electrode stack, wherein the electrode stack includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of separators; wherein each first electrode includes a first collector and a first cell tab, wherein each second electrode includes a second collector and a second cell tab, wherein a plurality of the first cell tabs extends from the electrode stack in a first longitudinal direction, and wherein a plurality of the second cell tabs extends from the electrode stack in a second longitudinal direction that is opposed to the first longitudinal direction. The process continues by forming a first conductive adhesive pad on an end cap, wherein the first conductive adhesive pad electrically couples to a first terminal arranged on the end cap; forming a second conductive adhesive pad on an enclosure, wherein the second conductive adhesive pad

electrically couples to a second terminal arranged on the enclosure; installing the end cap onto the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad; inserting the electrode stack into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad; and assembling the end cap onto the enclosure.

(6) Another aspect of the disclosure may include installing the end cap onto the electrode stack such that the plurality of the first cell tabs electrically connects to the first conductive adhesive pad.

(7) Another aspect of the disclosure may include inserting the electrode stack into the enclosure such that the plurality of the second cell tabs electrically connects to the second conductive adhesive pad.

(8) Another aspect of the disclosure may include assembling the electrode stack by forming the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators as planar sheets; and alternatingly stacking one of the plurality of first electrodes, and one the plurality of second electrodes with one of the plurality of separators interposed therebetween to form an intermediate stack, wherein the first cell tab extends outwardly therefrom in the first longitudinal direction, and wherein the second cell tab extends outwardly therefrom in the second longitudinal direction; and rolling the intermediate stack into a cylindrical shape.

(9) Another aspect of the disclosure may include curing the first conductive adhesive pad subsequent to installing the end cap onto the electrode stack with the plurality of the first cell tabs to electrically connect the first cell tabs to the first terminal arranged on the end cap and weldlessly mechanically join the first cell tabs to the end cap.

(10) Another aspect of the disclosure may include curing the second conductive adhesive pad subsequent to assembling the end cap onto the enclosure to electrically connect the second cell tabs to the enclosure and weldlessly mechanically join the second cell tabs to the second terminal arranged on the enclosure.

(11) Another aspect of the disclosure may include forming the first conductive adhesive pad on the end cap by applying a malleable curable conductive adhesive material onto a surface of the end cap.

(12) Another aspect of the disclosure may include the malleable curable conductive adhesive material being a thermosetting polymer.

(13) Another aspect of the disclosure may include forming the second conductive adhesive pad on the enclosure by applying a malleable curable conductive adhesive material onto an inner surface of the enclosure.

(14) Another aspect of the disclosure may include curing the first conductive adhesive pad by applying heat to the first conductive adhesive pad.

(15) Another aspect of the disclosure may include a method for assembling a battery cell, which includes assembling an electrode stack, wherein the electrode stack includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of separators; wherein each first electrode includes a first collector and a first cell tab, wherein each second electrode includes a second collector and a second cell tab, wherein a plurality of the first cell tabs extends from a first end of the electrode stack, and wherein a plurality of the second cell tabs extends from a second end of the electrode stack; forming a first conductive adhesive pad on a first end cap, wherein the first conductive adhesive pad electrically couples to a first terminal on the first end cap; installing the first end cap onto the first end of the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad; forming a second conductive adhesive pad on an enclosure, wherein the second conductive adhesive pad electrically couples to a second terminal arranged on the enclosure; inserting the electrode stack into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad; and assembling the end cap onto the enclosure.

(16) Another aspect of the disclosure may include installing the end cap onto the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad by installing

the end cap onto the electrode stack such that the plurality of the first cell tabs electrically connects to the first conductive adhesive pad.

(17) Another aspect of the disclosure may include inserting the electrode stack into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad by inserting the electrode stack into the enclosure such that the plurality of the second cell tabs electrically connects to the second conductive adhesive pad.

(18) Another aspect of the disclosure may include assembling the electrode stack by forming the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators as planar sheets; and alternatingly stacking one of the plurality of first electrodes and one the plurality of second electrodes with one of the plurality of separators interposed therebetween, wherein the first cell tab extends outwardly from the first end of the electrode stack, and wherein the second cell tab extends outwardly from the second end of the electrode stack.

(19) Another aspect of the disclosure may include curing the first conductive adhesive pad subsequent to installing the end cap onto the electrode stack with the plurality of the first cell tabs to electrically connect the first cell tabs to the first terminal arranged on the end cap and weldlessly mechanically join the first cell tabs to the end cap; and curing the second conductive adhesive pad subsequent to assembling the end cap onto the enclosure to electrically connect and weldlessly mechanically join the second cell tabs to the second terminal arranged on the enclosure.

(20) Another aspect of the disclosure may include a method for assembling a battery cell that includes assembling an electrode stack, wherein the electrode stack includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of separators; wherein each first electrode includes a first collector and a first cell tab, wherein each second electrode includes a second collector and a second cell tab, wherein a plurality of the first cell tabs extends from a first side of a first end of the electrode stack, and wherein a plurality of the second cell tabs extends from a second side of the first end of the electrode stack; forming a first conductive adhesive pad on a first side of an end cap, wherein the first conductive adhesive pad electrically couples to a first terminal on the end cap; forming a second conductive adhesive pad on a second side of the end cap, wherein the second conductive adhesive pad electrically couples to a second terminal on the end cap; installing the end cap onto the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad and the plurality of the second cell tabs engages the second conductive adhesive pad; inserting the electrode stack including the end cap into an enclosure; and assembling the end cap onto the enclosure.

(21) Another aspect of the disclosure may include installing the end cap onto the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad by installing the end cap onto the electrode stack such that the plurality of the first cell tabs electrically connects to the first conductive adhesive pad.

(22) Another aspect of the disclosure may include inserting the electrode stack into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad by installing the end cap onto the electrode stack such that the plurality of the second cell tabs electrically connects to the second conductive adhesive pad.

(23) Another aspect of the disclosure may include assembling the electrode stack by forming the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators as planar sheets; and alternatingly stacking one of the plurality of first electrodes, and one of the plurality of second electrodes with one of the plurality of separators interposed therebetween, wherein the first cell tab is arranged on the first side, and wherein the second cell tab is arranged on the second side.

(24) Another aspect of the disclosure may include curing the first conductive adhesive pad subsequent to installing the end cap onto the electrode stack with the plurality of the first cell tabs to electrically connect the first cell tabs to the first terminal arranged on the end cap and weldlessly mechanically join the first cell tabs to the end cap; and curing the second conductive adhesive pad

subsequent to installing the end cap onto the electrode stack with the plurality of the second cell tabs to electrically connect the second cell tabs to the second terminal arranged on the end cap and weldlessly mechanically join the second cell tabs to the end cap.

(25) Another aspect of the disclosure may include an electrochemical battery module that includes an electrode stack, wherein the electrode stack includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of separators; an end cap, the end cap having a first conductive adhesive pad, a first terminal, a second conductive adhesive pad, and a second terminal; and an enclosure. The electrode stack is arranged within the enclosure and the end cap; the first conductive adhesive pad electrically couples to the first terminal; the second conductive adhesive pad electrically couples to the second terminal; the plurality of first electrodes is electrically coupled to the first terminal via the first conductive adhesive pad; and the plurality of second electrodes is electrically coupled to the second terminal via the second conductive adhesive pad.

(26) Another aspect of the disclosure may include each of the plurality of first electrodes including a first collector and a first cell tab; each of the plurality of second electrodes including a second collector and a second cell tab; each of the first cell tabs of the plurality of first electrodes being electrically coupled to the first terminal and mechanically joined to the end cap via the first conductive adhesive pad; and each of the second cell tabs of the plurality of second electrodes being electrically coupled to the second terminal and mechanically joined to the end cap via the second conductive adhesive pad.

(27) Another aspect of the disclosure may include the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators being arranged as planar sheets; and the plurality of first electrodes and the plurality of second electrodes being alternatingly stacked with one of the plurality of separators interposed therebetween.

(28) Another aspect of the disclosure may include the enclosure being arranged as a cylindrical device; wherein the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators are arranged in a cylindrical shape and inserted into the enclosure.

(29) Another aspect of the disclosure may include the plurality of first electrodes being mechanically joined to the end cap via the first conductive adhesive pad; and the plurality of second electrodes being mechanically joined to the end cap via the second conductive adhesive pad.

(30) The above summary is not intended to represent every possible embodiment or every aspect of the present disclosure. Rather, the foregoing summary is intended to exemplify some of the novel aspects and features disclosed herein. The above features and advantages, and other features and advantages of the present disclosure, will be readily apparent from the following detailed description of representative embodiments and modes for carrying out the present disclosure when taken in connection with the accompanying drawings and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) One or more embodiments will now be described, by way of example, with reference to the accompanying drawings, in which:

(2) FIGS. **1-1** through **1-4** schematically illustrate aspects of a cutaway sideview of a cylindrical battery assembly, in accordance with the disclosure.

(3) FIG. **1-5** pictorially illustrates a cylindrical battery assembly, in accordance with the disclosure.

(4) FIGS. **2-1** through **2-4** schematically illustrate aspects of a cutaway sideview of an embodiment of a prismatic battery assembly, in accordance with the disclosure.

(5) FIG. **2-5** pictorially illustrates an embodiment of a prismatic battery assembly, in accordance with the disclosure.

(6) FIGS. **3-1** through **3-4** schematically illustrate aspects of a cutaway sideview of an embodiment of a prismatic battery assembly, in accordance with the disclosure.
(7) FIG. **3-5** pictorially illustrates an embodiment of a prismatic battery assembly, in accordance with the disclosure.
(8) FIGS. **4-1** through **4-5** schematically illustrate aspects of a cutaway sideview of an embodiment of a pouch battery assembly, in accordance with the disclosure.
(9) FIGS. **5-1** through **5-5** schematically illustrate aspects of a cutaway sideview of an embodiment of a pouch battery assembly, in accordance with the disclosure.
(10) The appended drawings are not necessarily to scale, and may present a somewhat simplified representation of various features of the present disclosure as disclosed herein, including, for example, specific dimensions, orientations, locations, and shapes. Details associated with such features will be determined in part by the particular intended application and use environment.

DETAILED DESCRIPTION

(11) The components of the disclosed embodiments, as described and illustrated herein, may be arranged and designed in a variety of different configurations. Thus, the following detailed description is not intended to limit the scope of the disclosure, but is merely representative of possible embodiments thereof. In addition, while numerous specific details are set forth in the following description in order to provide a thorough understanding of the embodiments disclosed herein, some embodiments may be practiced without some of these details. Moreover, for the purpose of clarity, certain technical material that is understood in the related art has not been described in detail to avoid unnecessarily obscuring the disclosure.
(12) Furthermore, the drawings are in simplified form and are not to precise scale. For purposes of convenience and clarity, directional terms such as top, bottom, left, right, up, over, above, below, beneath, rear, and front, may be used with respect to the drawings. These and similar directional terms are not to be construed to limit the scope of the disclosure. Furthermore, the disclosure, as illustrated and described herein, may be practiced in the absence of an element that is not specifically disclosed herein.
(13) The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by an expressed or implied theory presented herein. Throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.
(14) For the sake of brevity, some components and techniques and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. Many alternative or additional functional relationships or physical connections may be present in an embodiment of the disclosure.
(15) FIGS. **1-1** through **1-5** illustrate details related to an embodiment of a cylindrical battery cell (or battery module) **100** and a related assembly process, wherein positive and negative terminals are electrically connected and weldlessly, mechanically joined to an electrode stack employing an electrically conductive mechanically adhesive material. The battery cell **100** includes an electrode stack **110** that is contained with an enclosure **140** and an end cap **130**. The enclosure **140** includes and defines a first terminal **142**, and the end cap **130** includes a second terminal **132**. When assembled, the electrode stack **110** is sealed within the enclosure **140** using the end cap **130**. In one embodiment, the end cap **130** includes an electrical isolator **134** that electrically isolates the second terminal **132** from the end cap **130**, as illustrated with reference to FIG. **1-4**.
(16) As employed herein, the terms “seal”, “sealed”, and related terms refer to fastening or attaching the joined elements or members such that the sealed portion prevents passage of liquids, gases, solids, gelids, etc., therethrough.
(17) The electrode stack **110** has a plurality of alternating first and second electrodes **111**, **121**,

respectively, that are arranged with separators **115** interposed therebetween. In one non-limiting embodiment, and as shown, the first electrode **111** is an anode, and the second electrode **121** is a cathode. Each first electrode **111** includes a first collector **112** and a first cell tab **114** that projects outwardly in a longitudinal direction **105** from a first end **101** of the electrode stack **110**. Each second electrode **121** includes a second collector **122** and a second cell tab **124** that projects outwardly in the longitudinal direction **105** from a second end **102** of the electrode stack **110**.

(18) The electrode stack **110** is created by forming each of the first electrodes **111**, the second electrodes **121**, and the separators **115** as planar sheets, and alternatingly stacking the first electrodes **111** and the second electrodes **121** with one of the separators **115** interposed therebetween to form an intermediate stack (not shown). The first cell tabs **114** project outwardly in the longitudinal direction **105** from the first end **101** of the electrode stack **110** and the second cell tabs **124** project outwardly in the longitudinal direction **105** from the second end **102** of the electrode stack **110**. The intermediate stack is wound or rolled into a cylindrical shape for further processing and assembly.

(19) In one embodiment, the first electrodes **111** are composed as anodes that include a lithium host material such as graphite, silicon, or lithium titanate that is arranged on the first collector **112**, which is a copper or copper alloy current collector. In one embodiment, the second electrodes **121** are composed as cathodes that include lithium-based active material that stores intercalated lithium at a higher electrochemical potential than the lithium host material used to make the anode, and is arranged on the second collector **122**, which is an aluminum or aluminum alloy current collector. The separators **115** are each composed of one or more porous polymer layers that individually may be composed of one or a wide variety of polymers. Each of the one or more polymer layers may be a polyolefin. The polymer layer(s) function to electrically insulate and physically separate the first and second electrodes **111**, **121**.

(20) The second end **102** of the electrode stack **110** is assembled onto the end cap **130** so that the second cell tabs **124** of the second electrodes **121** electrically connect to the second terminal **132** of the end cap **130**, and weldlessly mechanically join the second cell tabs **124** to the end cap **130** via a second conductive adhesive pad **135**.

(21) The second conductive adhesive pad **135** is a malleable curable electrically conductive adhesive material.

(22) The malleable curable electrically conductive adhesive material may be fabricated by loading an adhesive component with conductive fillers. Non-limiting examples of adhesive materials include organic or synthetic resins, varnish, or silicone. Non-limiting examples of conductive fillers include metals, metal oxides, or metal nitrides such as aluminum, copper, silver, nickel, and graphite in their various forms.

(23) The second conductive adhesive pad **135** is formed from a conductive adhesive material that may be cured by various methods such as heat cure, air drying cure, moisture cure, radiation cure or using a hardener agent, e.g., a 2-part adhesive. In one embodiment, the malleable curable conductive adhesive material includes a thermosetting polymer that is curable in place without additional processing steps.

(24) The malleable curable conductive adhesive is applied onto a surface of the end cap in an uncured state to form the second conductive adhesive pad **135**, and is subjected to a curing process after assembly of the battery cell **100**.

(25) When the malleable curable conductive adhesive is a thermosetting polymer, the curing process may require the addition of heat to effect curing.

(26) Subsequent to assembling the second end **102** of the electrode stack **110** onto the end cap **130**, the first end **101** of the electrode stack **110** is inserted into the enclosure **140**. This is schematically illustrated with reference to FIG. **1-2**. The enclosure **140** is arranged as a cylindrical device fabricated from a conductive polymer or other material, and has an open first end **141** and a closed second end **143**. All or a portion of the enclosure **140** may function as the first terminal **142**.

(27) The malleable curable conductive adhesive is applied onto an interior portion of the second end **143** of the enclosure **140** in an uncured state to form a first conductive adhesive pad **145**, and electrically connects to the first terminal **142**. The malleable curable conductive adhesive of the first conductive adhesive pad **145** is subjected to a curing process after assembly of the battery cell **100**.
(28) The end cap **130** is sealably attached to the enclosure **140** to form the battery cell **100**, as illustrated with reference to FIG. **1-3**, with an embodiment of the battery cell **100** being pictorially illustrated with reference to FIG. **1-5**.
(29) This arrangement for assembling the cylindrical battery cell **100** enables the electrode stack **110** to be wound without notching or folding, thus permitting it to be inserted directly into the enclosure **140** with a pre-applied conductive adhesive and absent interconnects. The end cap **130** may be adhered directly to the wound electrode stack **110** with conductive adhesive.
(30) FIGS. **2-1** through **2-5** illustrate details related to an embodiment of a prismatic N-type battery cell (or battery module) **200** and a related assembly process, wherein positive and negative terminals are electrically connected and weldlessly, mechanically joined to an electrode stack employing an electrically conductive mechanically adhesive material. The battery cell **200** includes an electrode stack **210** that is contained with an enclosure **240** and an end cap **230**. The enclosure **240** includes and defines a first terminal **242**, and the end cap **230** includes a second terminal **232**. When assembled, the electrode stack **210** is sealed within the enclosure **240** using the end cap **230**. In one embodiment, the end cap **230** includes an electrical isolator **234** that electrically isolates the second terminal **232** from the end cap **230**, as illustrated with reference to FIG. **2-4**.
(31) The electrode stack **210** has a plurality of alternating first and second electrodes **211**, **221**, respectively, that are arranged with separators **215** interposed therebetween. In one non-limiting embodiment, and as shown, the first electrode **211** is an anode, and the second electrode **221** is a cathode. Each first electrode **211** includes a first collector **212** and a first cell tab **214** that projects outwardly in a longitudinal direction **205** from a first end **201** of the electrode stack **210**. Each second electrode **221** includes a second collector **222** and a second cell tab **224** that projects outwardly in the longitudinal direction **205** from a second end **202** of the electrode stack **210**.
(32) The electrode stack **210** is created by forming each of the first electrodes **211**, the second electrodes **221**, and the separators **215** as planar sheets, and alternatingly stacking the first electrodes **211** and the second electrodes **221** with one of the separators **215** interposed therebetween to form an intermediate stack (not shown). The first cell tabs **214** project outwardly in the longitudinal direction **205** from the first end **201** of the electrode stack **210** and the second cell tabs **224** project outwardly in the longitudinal direction **205** from the second end **202** of the electrode stack **210**.
(33) In one embodiment, the first electrodes **211** are composed as anodes that include a lithium host material such as graphite, silicon, or lithium titanate that is arranged on the first collector **212**, which is a copper or copper alloy current collector. In one embodiment, the second electrodes **221** are composed as cathodes that include lithium-based active material that stores intercalated lithium at a higher electrochemical potential than the lithium host material used to make the anode, and is arranged on the second collector **222**, which is an aluminum or aluminum alloy current collector. The separators **215** are each composed as one or more porous polymer layers that individually may be composed of one or a wide variety of polymers. Each of the one or more polymer layers may be a polyolefin. The polymer layer(s) function to electrically insulate and physically separate the first and second electrodes **211**, **221**.
(34) The second end **202** of the electrode stack **210** is assembled onto the end cap **230** so that the second cell tabs **224** of the second electrodes **221** electrically connect to the second terminal **232** of the end cap **230**, and weldlessly mechanically join the second cell tabs **224** to the end cap **230** via a second conductive adhesive pad **235**.
(35) The second conductive adhesive pad **235** is a malleable curable conductive adhesive material.

The malleable curable conductive adhesive is applied onto a surface of the end cap in an uncured state to form the second conductive adhesive pad **235**, and is subjected to a curing process after assembly of the battery cell **200**.
(36) Subsequent to assembling the second end **202** of the electrode stack **210** onto the end cap **230**, the first end **201** of the electrode stack **210** is inserted into the enclosure **240**. This is schematically illustrated with reference to FIG. **2-2**. The enclosure **240** is arranged as a rectangular prism device fabricated from a conductive polymer or other material, and has an open first end **241** and a closed second end **243**. All or a portion of the enclosure **240** may function as the first terminal **242**.
(37) The malleable curable conductive adhesive is applied onto an interior portion of the second end **243** of the enclosure **240** in an uncured state to form the first conductive adhesive pad **245**, and electrically connects to the first terminal **242**. The malleable curable conductive adhesive is subjected to a curing process after assembly of the battery cell **200**.
(38) The end cap **230** is sealably attached to the enclosure **240** to form the battery cell **200**, as illustrated with reference to FIG. **2-3**, with embodiments of the battery cell **200** being pictorially illustrated with reference to FIG. **2-5**.
(39) This arrangement for assembling the prismatic N-type battery cell **200** enables the electrode stack **210** to be assembled without notching or folding, thus permitting it to be inserted directly into the enclosure **240** with a pre-applied conductive adhesive and without interconnects. The end cap **230** may be adhered directly to the electrode stack **210** with conductive adhesive.
(40) FIGS. **3-1** through **3-5** illustrate details related to an embodiment of a prismatic P-type battery cell (or battery module) **300** and a related assembly process, wherein positive and negative terminals are electrically connected and weldlessly, mechanically joined to an electrode stack employing an electrically conductive mechanically adhesive material. The battery cell **300** includes an electrode stack **310** that is contained with an enclosure **340** and an end cap **330**. The end cap **330** includes a first terminal **342** and a second terminal **332**. When assembled, the electrode stack **310** is sealed within the enclosure **340** using the end cap **330**.
(41) FIG. **3-1** schematically illustrates a cutaway sideview of electrode stack **310**, and FIG. **3-2** schematically illustrates a corresponding cutaway end view of the electrode stack **310**.
(42) The electrode stack **310** has a plurality of alternating first and second electrodes **311**, **321**, respectively, that are arranged with separators **315** interposed therebetween. The electrode stack **310** includes a first end **301** and a second end **302**. In one non-limiting embodiment, and as shown, the first electrode **311** is an anode, and the second electrode **321** is a cathode. Each first electrode **311** includes a first collector **312** and a first cell tab **314** that projects outwardly in a longitudinal direction **305** from the second end **302** of the electrode stack **310**. Each second electrode **321** includes a second collector **322** and a second cell tab **324** that projects outwardly in the longitudinal direction **305** from the second end **302** of the electrode stack **310**. As shown, the first electrode **311** is arranged on a leftward side, and the second electrode **321** is arranged on a rightward side.
(43) The electrode stack **310** is created by forming each of the first electrodes **311**, the second electrodes **321**, and the separators **315** as planar sheets, and alternatingly stacking the first electrodes **311** and the second electrodes **321** with one of the separators **315** interposed therebetween to form an intermediate stack.
(44) In one embodiment, the first electrodes **311** are composed as anodes that include a lithium host material such as graphite, silicon, or lithium titanate that is arranged on the first collector **312**, which is a copper or copper alloy current collector. In one embodiment, the second electrodes **321** are composed as cathodes that include lithium-based active material that stores intercalated lithium at a higher electrochemical potential than the lithium host material used to make the anode, and is arranged on the second collector **322**, which is an aluminum or aluminum alloy current collector. The separators **315** are each composed of one or more porous polymer layers that individually may be composed of one or a wide variety of polymers. Each of the one or more polymer layers may be

a polyolefin. The polymer layer(s) functions to electrically insulate and physically separate the first and second electrodes **311**, **321**.
(45) The second end **302** of the electrode stack **310** is assembled onto the end cap **330**.
(46) This includes the first cell tabs **314** of the first electrodes **311** being electrically connected to the first terminal **342** of the end cap **330**, and being weldlessly mechanically joined to the end cap **330** via a first conductive adhesive pad **345**.
(47) This includes the second cell tabs **324** of the second electrodes **321** being electrically connected to the second terminal **332** of the end cap **330**, and being weldlessly mechanically joined to the end cap **330** via a second conductive adhesive pad **335**.
(48) The first and second conductive adhesive pads **345**, **335** are formed from a malleable curable conductive adhesive material. The malleable curable conductive adhesive is applied onto a surface of the end cap in an uncured state to form the second conductive adhesive pad **335**, and is subjected to a curing process after assembly of the battery cell **300**.
(49) Subsequent to assembling the second end **302** of the electrode stack **310** onto the end cap **330**, the first end **301** of the electrode stack **310** is inserted into the enclosure **340**. This is schematically illustrated with reference to FIG. **3-3**. The enclosure **340** is arranged as a rectangular prism device fabricated from a conductive polymer or other material, and has an open first end **341** and a closed second end **343**.
(50) The malleable curable conductive adhesive is applied onto an interior portion of the second end **343** of the enclosure **340** in an uncured state to form the first conductive adhesive pad **345**, and electrically connects to the first terminal **342**. The malleable curable conductive adhesive is subjected to a curing process after assembly of the battery cell **300**.
(51) The end cap **330** is sealably attached to the enclosure **340** to form the battery cell **300**, as illustrated with reference to FIG. **3-3**, with embodiments of the battery cell **300** being pictorially illustrated with reference to FIG. **3-5**.
(52) FIGS. **4-1** through **4-5** illustrate details related to an embodiment of a pouch N-type battery cell (or battery module) **400** and a related assembly process, wherein positive and negative terminals are electrically connected and weldlessly, mechanically joined to an electrode stack employing an electrically conductive mechanically adhesive material. The battery cell **400** includes an electrode stack **410** that is contained with an enclosure **440**, with a first end cap **45** and a second end cap **430**.
(53) The first end cap **450** includes and defines a first terminal **442**, and the second end cap **430** includes a second terminal **432**. When assembled, the electrode stack **410** is sealed within the enclosure **440** using the first and second end caps **450**, **430**. FIG. **4-4** schematically illustrates a cutaway top view of the pouch N-type battery cell **400** including the first end cap **450**A.
(54) The electrode stack **410** has a plurality of alternating first and second electrodes **411**, **421**, respectively, that are arranged with separators **415** interposed therebetween. In one non-limiting embodiment, and as shown, the first electrode **411** is an anode, and the second electrode **421** is a cathode. Each first electrode **411** includes a first collector **412** and a first cell tab **414** that projects outwardly in a longitudinal direction **405** from a first end **401** of the electrode stack **410**. Each second electrode **421** includes a second collector **422** and a second cell tab **424** that projects outwardly in the longitudinal direction **405** from a second end **402** of the electrode stack **410**.
(55) The electrode stack **410** is created by forming each of the first electrodes **411**, the second electrodes **421**, and the separators **415** as planar sheets, and alternatingly stacking the first electrodes **411** and the second electrodes **421** with one of the separators **415** interposed therebetween to form an intermediate stack. The first cell tabs **414** project outwardly in the longitudinal direction **405** from the first end **401** of the electrode stack **410** and the second cell tabs **424** project outwardly in the longitudinal direction **405** from the second end **402** of the electrode stack **410**.
(56) In one embodiment, the first electrodes **411** are composed as anodes that include a lithium host

material such as graphite, silicon, or lithium titanate that is arranged on the first collector **412**, which is a copper or copper alloy current collector. In one embodiment, the second electrodes **421** are composed as cathodes that include lithium-based active material that stores intercalated lithium at a higher electrochemical potential than the lithium host material used to make the anode, and is arranged on the second collector **422**, which is an aluminum or aluminum alloy current collector. The separators **415** are each composed of one or more porous polymer layers that individually may be composed of one or a wide variety of polymers. Each of the one or more polymer layers may be a polyolefin. The polymer layer(s) function to electrically insulate and physically separate the first and second electrodes **411**, **421**.

(57) The first end **401** of the electrode stack **410** is assembled onto the first end cap **450** so that the first cell tabs **414** of the first electrodes **411** electrically connect to the first terminal **442** of the first end cap **450**, and weldlessly mechanically join the first cell tabs **414** to the first end cap **450** via a first conductive adhesive pad **445**.

(58) The second end **402** of the electrode stack **410** is assembled onto the second end cap **430** so that the second cell tabs **424** of the second electrodes **421** electrically connect to the second terminal **432** of the second end cap **430**, and weldlessly mechanically join the second cell tabs **424** to second the end cap **430** via a second conductive adhesive pad **435**.

(59) The first and second conductive adhesive pads **445**, **435** are formed from a malleable curable conductive adhesive material.

(60) Subsequent to assembling the second end **402** of the electrode stack **410** onto the end cap **430**, the first end **401** of the electrode stack **410** is inserted into the enclosure **440**. This is schematically illustrated with reference to FIG. **4-2**. The enclosure **440** is arranged as a rectangular prism device fabricated from a conductive polymer or other material, and has an open first end **441** and a closed second end **443**. All or a portion of the enclosure **440** may function as the first terminal **442**.

(61) The malleable curable conductive adhesive is applied onto an interior portion of the second end **443** of the enclosure **440** in an uncured state to form the first conductive adhesive pad **445**, and electrically connects to the first terminal **442**. The malleable curable conductive adhesive is subjected to a curing process after assembly of the battery cell **400**.

(62) The first and second end caps **450**, **430** are sealably attached to the enclosure **440** to form the battery cell **400**, as illustrated with reference to FIG. **4-3**, with embodiments of the battery cell **400** being pictorially illustrated with reference to FIG. **4-5**.

(63) This arrangement for assembling the pouch N-type battery cell **400** enables the electrode stack **410** to be assembled without notching or folding, thus permitting it to be inserted directly into the enclosure **440** with a pre-applied conductive adhesive and without interconnects. The end cap **430** may be adhered directly to the electrode stack **410** with conductive adhesive.

(64) FIGS. **5-1** through **5-5** illustrate details related to an embodiment of a pouch P-type battery cell (or battery module) **500** and a related assembly process, wherein positive and negative terminals are electrically connected and weldlessly, mechanically joined to an electrode stack employing an electrically conductive mechanically adhesive material. The battery cell **500** includes an electrode stack **510** that is contained with an enclosure **540** and an end cap **530**. The end cap **530** includes a first terminal **542** and a second terminal **532**. When assembled, the electrode stack **510** is sealed within the enclosure **540** using the end cap **530**. The end cap **530** may be assembled onto the enclosure **540** employing overmolding or using an adhesive material.

(65) FIG. **5-1** schematically illustrates a cutaway sideview of electrode stack **510**, and FIG. **5-2** schematically illustrates a corresponding cutaway end view of the electrode stack **510**.

(66) The electrode stack **510** has a plurality of alternating first and second electrodes **511**, **521**, respectively, that are arranged with separators **515** interposed therebetween. The electrode stack **510** includes a first end **501** and a second end **502**. In one non-limiting embodiment, and as shown, the first electrode **511** is an anode, and the second electrode **521** is a cathode. Each first electrode **511** includes a first collector **512** and a first cell tab **514** that projects outwardly in a longitudinal

direction **505** from the second end **502** of the electrode stack **510**. Each second electrode **521** includes a second collector **522** and a second cell tab **524** that projects outwardly in the longitudinal direction **505** from the second end **502** of the electrode stack **510**. As shown with reference to FIGS. **5-4** and **5-5**, the first electrode **511** is arranged on a leftward side of the end cap **530**, and the second electrode **521** is arranged on a rightward side of the end cap **530**.

(67) The electrode stack **510** is created by forming each of the first electrodes **511**, the second electrodes **521**, and the separators **515** as planar sheets, and alternatingly stacking the first electrodes **511** and the second electrodes **521** with one of the separators **515** interposed therebetween to form an intermediate stack.

(68) In one embodiment, the first electrodes **511** are composed as anodes that include a lithium host material such as graphite, silicon, or lithium titanate that is arranged on the first collector **512**, which is a copper or copper alloy current collector. In one embodiment, the second electrodes **521** are composed as cathodes that include lithium-based active material that stores intercalated lithium at a higher electrochemical potential than the lithium host material used to make the anode, and is arranged on the second collector **522**, which is an aluminum or aluminum alloy current collector. The separators **515** are each composed as one or more porous polymer layers that individually may be composed of one or a wide variety of polymers. Each of the one or more polymer layers may be a polyolefin. The polymer layer(s) function to electrically insulate and physically separate the first and second electrodes **511**, **521**.

(69) The second end **502** of the electrode stack **510** is assembled onto the end cap **530**.

(70) This includes the first cell tabs **514** of the first electrodes **511** being electrically connected to the first terminal **542** of the end cap **530**, and being weldlessly mechanically joined to the end cap **530** via a first conductive adhesive pad **545**.

(71) This includes the second cell tabs **524** of the second electrodes **521** being electrically connected to the second terminal **532** of the end cap **530**, and being weldlessly mechanically joined to the end cap **530** via a second conductive adhesive pad **535**.

(72) Each of the first and second conductive adhesive pads **545**, **535** is a malleable curable conductive adhesive material. The malleable curable conductive adhesive is applied onto a surface of the end cap in an uncured state to form the second conductive adhesive pad **535**, and is subjected to a curing process after assembly of the battery cell **500**.

(73) Subsequent to assembling the second end **502** of the electrode stack **510** onto the end cap **530**, the first end **501** of the electrode stack **510** is inserted into the enclosure **540**. This is schematically illustrated with reference to FIG. **5-3**. The enclosure **540** is arranged as a rectangular prism device fabricated from a conductive polymer or other material, and has an open first end **541** and a closed second end **543**.

(74) The malleable curable conductive adhesive is applied onto an interior portion of the second end **543** of the enclosure **540** in an uncured state to form the first conductive adhesive pad **545**, and electrically connects to the first terminal **542**. The malleable curable conductive adhesive is subjected to a curing process after assembly of the battery cell **500**.

(75) The end cap **530** is sealably attached to the enclosure **540** to form the battery cell **500**, as illustrated with reference to FIG. **5-3**, with embodiments of the battery cell **500** being pictorially illustrated with reference to FIG. **5-5**.

(76) Curing the malleable curable conductive adhesive includes, by way of non-limiting examples, snap cure, heat cure, room-temperature cure, or use of thermoset material or dual component conductive adhesives. Furthermore, the conductive adhesive pads described herein may be formed in place by deposition, or may be inserted after having been formed by ejection molding, 3D printing, rotational molding, injection molding, thermoforming, compression molding, blow molding, reaction injection molding, vacuum casting, resin casting.

(77) As employed herein, the term “weldless”, “weldlessly joined”, “weld-free”, and related terms refer to a process of joining two or more elements and a resulting junction that is formed by

adhesion of the elements, and without employing heat, electrical energy, or pressure in a manner that fuses the materials of the two or more elements to form a continuum.

(78) The concepts described herein may advantageously reduce or eliminate certain cell components that may be found on battery cells that employ welding processes as part of assembling battery cells, and the manufacturing processes. The concepts described herein may advantageously reduce or eliminate tears in foils of pouches on some embodiments. The concepts described herein may advantageously eliminate a need for external tabs on certain embodiments. The concepts described herein may advantageously preclude a need for welding processes, and thus eliminate welding-related risks such as burn-throughs, spatter, etc. The concepts described herein may advantageously facilitate electrical and mechanical connections between dissimilar metals while minimizing or eliminating risks of electrolysis and galvanic corrosion on certain embodiments. The concepts described herein may advantageously facilitate thermal conductivity to the electrodes on certain embodiments.

(79) The detailed description and the drawings or figures are supportive and descriptive of the present teachings, but the scope of the present teachings is defined solely by the claims. While some of the best modes and other embodiments for carrying out the present teachings have been described in detail, various alternative designs and embodiments exist for practicing the present teachings defined in the claims.

Claims

1. A method for assembling a battery cell, the method comprising: assembling an electrode stack, wherein the electrode stack includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of separators; wherein each first electrode includes a first collector and a first cell tab, wherein each second electrode includes a second collector and a second cell tab, wherein a plurality of the first cell tabs extends from the electrode stack in a first longitudinal direction, and wherein a plurality of the second cell tabs extends from the electrode stack in a second longitudinal direction that is opposed to the first longitudinal direction; forming a first conductive adhesive pad on an end cap, wherein the first conductive adhesive pad electrically couples to a first terminal arranged on the end cap; forming a second conductive adhesive pad on an enclosure, wherein the second conductive adhesive pad electrically couples to a second terminal arranged on the enclosure; installing the end cap onto the electrode stack such that the plurality of the second cell tabs engages the second conductive adhesive pad; inserting the electrode stack into the enclosure such that the plurality of the first cell tabs engages the first conductive adhesive pad; and assembling the end cap onto the enclosure.

2. The method of claim 1, wherein installing the end cap onto the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad comprises installing the end cap onto the electrode stack such that the plurality of the first cell tabs electrically connects to the first conductive adhesive pad.

3. The method of claim 1, wherein inserting the electrode stack into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad comprises inserting the electrode stack into the enclosure such that the plurality of the second cell tabs electrically connects to the second conductive adhesive pad.

4. The method of claim 1, wherein assembling the electrode stack comprises: forming the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators as planar sheets; alternatingly stacking one of the plurality of first electrodes, and one the plurality of second electrodes with one of the plurality of separators interposed therebetween to form an intermediate stack, wherein the first cell tab extends outwardly therefrom in the first longitudinal direction, and wherein the second cell tab extends outwardly therefrom in the second longitudinal direction; and rolling the intermediate stack into a cylindrical shape.

5. The method of claim 1, further comprising curing the first conductive adhesive pad subsequent to installing the end cap onto the electrode stack with the plurality of the first cell tabs to electrically connect the first cell tabs to the first terminal arranged on the end cap and weldlessly mechanically join the first cell tabs to the end cap.
6. The method of claim 1, further comprising curing the second conductive adhesive pad subsequent to assembling the end cap onto the enclosure to electrically connect the second cell tabs to the enclosure and weldlessly mechanically join the second cell tabs to the second terminal arranged on the enclosure.
7. The method of claim 1, wherein forming the first conductive adhesive pad on the end cap comprises applying a malleable curable conductive adhesive material onto a surface of the end cap.
8. The method of claim 7, wherein the malleable curable conductive adhesive material comprises a thermosetting polymer.
9. The method of claim 1, wherein forming the second conductive adhesive pad on the enclosure comprises applying a malleable curable conductive adhesive material onto an inner surface of the enclosure.
10. The method of claim 9, wherein curing the first conductive adhesive pad comprises applying heat to the first conductive adhesive pad.
11. A method for assembling a battery cell, the method comprising: assembling an electrode stack, wherein the electrode stack includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of separators; wherein each first electrode includes a first collector and a first cell tab, wherein each second electrode includes a second collector and a second cell tab, wherein a plurality of the first cell tabs extends from a first end of the electrode stack, and wherein a plurality of the second cell tabs extends from a second end of the electrode stack; forming a first conductive adhesive pad on a first end cap, wherein the first conductive adhesive pad electrically couples to a first terminal on the first end cap; installing the first end cap onto the first end of the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad; forming a second conductive adhesive pad on an enclosure, wherein the second conductive adhesive pad electrically couples to a second terminal arranged on the enclosure; inserting the electrode stack into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad; and assembling the end cap onto the enclosure.
12. The method of claim 11, wherein installing the end cap onto the electrode stack such that the plurality of the first cell tabs engages the first conductive adhesive pad comprises installing the end cap onto the electrode stack such that the plurality of the first cell tabs electrically connects to the first conductive adhesive pad.
13. The method of claim 11, wherein inserting the electrode stack into the enclosure such that the plurality of the second cell tabs engages the second conductive adhesive pad comprises inserting the electrode stack into the enclosure such that the plurality of the second cell tabs electrically connects to the second conductive adhesive pad.
14. The method of claim 11, wherein assembling the electrode stack comprises: forming the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators as planar sheets; and alternatingly stacking one of the plurality of first electrodes and one the plurality of second electrodes with one of the plurality of separators interposed therebetween, wherein the first cell tab extends outwardly from the first end of the electrode stack, and wherein the second cell tab extends outwardly from the second end of the electrode stack.
15. The method of claim 11, further comprising: curing the first conductive adhesive pad subsequent to installing the end cap onto the electrode stack with the plurality of the first cell tabs to electrically connect the first cell tabs to the first terminal arranged on the end cap and weldlessly mechanically join the first cell tabs to the end cap; and curing the second conductive adhesive pad subsequent to assembling the end cap onto the enclosure to electrically connect and weldlessly mechanically join the second cell tabs to the second terminal arranged on the enclosure.

16. An electrochemical battery module, comprising: an electrode stack, wherein the electrode stack includes a plurality of first electrodes, a plurality of second electrodes, and a plurality of separators; an end cap, the end cap having a first conductive adhesive pad, and a first terminal; and an enclosure the enclosure having a second conductive adhesive pad, and a second terminal; wherein the electrode stack is arranged within the enclosure and the end cap; wherein the first conductive adhesive pad electrically couples to the first terminal; wherein the second conductive adhesive pad electrically couples to the second terminal; wherein the plurality of first electrodes is electrically coupled to the first terminal via the first conductive adhesive pad; and wherein the plurality of second electrodes is electrically coupled to the second terminal via the second conductive adhesive pad.

17. The electrochemical battery module of claim 16: wherein each of the plurality of first electrodes includes a first collector and a first cell tab; wherein each of the plurality of second electrodes includes a second collector and a second cell tab; wherein each of the first cell tabs of the plurality of first electrodes is electrically coupled to the first terminal and mechanically joined to the end cap via the first conductive adhesive pad; and wherein each of the second cell tabs of the plurality of second electrodes is electrically coupled to the second terminal and mechanically joined to the end cap via the second conductive adhesive pad.

18. The electrochemical battery module of claim 16: wherein the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators are arranged as planar sheets; and wherein the plurality of first electrodes and the plurality of second electrodes are alternatingly stacked with one of the plurality of separators interposed therebetween.

19. The electrochemical battery module of claim 18: further comprising: the enclosure being arranged as a cylindrical device; wherein the plurality of first electrodes, the plurality of second electrodes, and the plurality of separators are arranged in a cylindrical shape and inserted into the enclosure.

20. The electrochemical battery module of claim 16, further comprising: wherein the plurality of first electrodes is mechanically joined to the end cap via the first conductive adhesive pad; and wherein the plurality of second electrodes is mechanically joined to the end cap via the second conductive adhesive pad.